THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET D

Chapters 5 & 6: **Number Play, We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number is divisible by 8 if its last three digits form a number divisible by: 1

(a) 2 **(b)** 5 **(c)** 8 **(d)** 4

2. The sum of any two odd numbers is: 1

(a) prime **(b)** a multiple of 3 **(c)** even **(d)** odd

3. A palindromic number reads: 1

(a) as a prime **(b)** the same forwards and backwards

(c) only forwards **(d)** differently backwards

4. If a number is divisible by both 4 and 9, it is also divisible by: 1

(a) 13 **(b)** 36 **(c)** 18 only **(d)** 27

5. Which of the following numbers is a perfect number? 1

(a) 36 **(b)** 30 **(c)** 20 **(d)** 28

6. $(a+b)(a-b)$ equals: 1

(a) a^2+b^2 **(b)** $(a-b)^2$ **(c)** $a^2-2ab+b^2$ **(d)** a^2-b^2

7. The multiplicative identity is: 1

(a) 1 **(b)** 0 **(c)** 10 **(d)** −1

8. The value of $(x+1)^2$ when $x=9$ is: 1

(a) 91 **(b)** 81 **(c)** 100 **(d)** 99

9. $-3\times(4-6)$ equals: 1

(a) 18 **(b)** 6 **(c)** −30 **(d)** −6

10. The number of terms in the expansion of $(x+y)(p+q)$ is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the remainder when 245 is divided by 9 using digit sum. 2

12. Is a number divisible by 12 if it is divisible by 3 and 4? Give a reason. 2

13. Why is 0 called the additive identity? 2

14. Expand $(m+n)(m-n)$ and give the name of the identity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Is 20000001 divisible by 11? Show the working. **3**

16. Expand $(3m - 2n)^2$ and find its value when $m = 2$, $n = 1$. **3**

17. Show that a number and the sum of its digits leave the same remainder on division by 9. Verify for 4729. **3**

OR

Write the first five multiples of 14 and say which are divisible by 4. **3**

18. A rectangle has length $(x + 5)$ and breadth $(x - 2)$. Find its area as an expression and at $x = 6$. **3**

OR

Show by an example that $(a - b)^2 \neq a^2 - b^2$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) What are twin primes? List the pairs below 50.
(ii) Is 91 prime? **5**

OR

Case study: A school bell rings 4 times during an hour at equal gaps of 15 minutes starting at 8:00. **5**
(i) What are the ring times?
(ii) What is the LCM-based interpretation if another system rings every 20 minutes?
(iii) When do they ring together?

20. (i) Show $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ with $a = 1, b = 2, c = 3$. **5**
(ii) Evaluate $(10 + 2 + 1)^2$.

OR

Case study: A farmer has a square field of side 20 m. He extends one side by 3 m and shortens the other by 3 m. **5**
(i) New area?
(ii) Compare with the original.
(iii) What identity explains?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND